Project Proposal: Evaluating LLM Performance on Medical Question across Clinical Specialities

Srijan Ray, Evan Fix, Vijay Damineni

Project Overview

We will be using the PubMedQA (https://huggingface.co/datasets/qiaojin/PubMedQA) Dataset. The overall goal of this project is to evaluate LLM performance on different clinical question types and identify if the LLMs have any systemic biases regarding certain questions; for example, if the LLMs constantly struggle with questions in the pharmacology or genetics domains.

Dataset

This dataset is used to answer a variety of yes or no biomedical questions. The dataset gives sufficient context on what was used to answer every question. For every question, it provides both a short answer yes or no and a long answer with more explanation.

Research Question

- How differently do LLMs perform across different specialties in MedQA?
- What are their systematic blind spots?
- How does chain-of-thought processing affect LLM performance across different specialty difficulties?

Methodology

- Which LLMs are we using, need architectural differences to draw sufficient conclusions. Currently thinking: LLaMA 4, Claude 3, and Gemma 7B or newer versions of these models.
- How we are going to analyze their performance (precision, recall, f1-score, confusion matrices)
- Majority-class vs random baselines
- Zero-shot vs few-shot vs chain-of-thought comparisons

Team Member Responsibilities

1. Person 1: Collect Dataset, clean dataset for injection into both free and open source LLM models (Gemma, LLaMa, Qwen) and closed-source models (GPT-5.4, Sonnet 4.6, Gemini 2.5 Pro).

2. Person 2: Run LLMs on the dataset cleaned and collected by Person 1, run models using zero-shot, few-shot, and CoT, record results per model
3. Person 3: Analyze results, create visualizations on model performance, write up report

For each of the team member responsibilities listed above, the other two team members will provide support for the team member in their stage of the project. For example, if Person 2 is running the LLMs on the dataset, Person 1 and Person 3 will still help in assisting Person 2 if they encounter any issues and also assist in recording/tabling the results. This step ensures equal distribution of responsibilities.

Originality and Significance

While there is a lot of research done on whether LLMs can answer biomedical questions, there hasn't been a lot of emphasis on what domains of biomedical research, such as cardiology, genetics … etc, that LLMs struggle with. Our project is unique, because we focus on what domains in biomedical research LLMs struggle with and how different prompting strategies would help with improving reasoning.

This project is significant, because a variety of people from doctors to citizens may choose to rely on LLMs for biomedical advice. We need to know if LLMS fail in specific domains or if there are any systematic blind spots. This way we can see where additional safeguards are needed with using LLMS for biomedical advice.